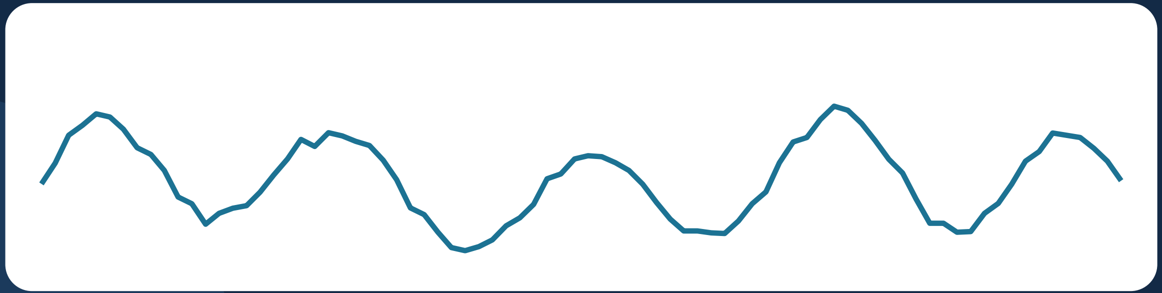
SESSION 5

Artifacts in EEG

Sources and Visual Identification

"NO DATA IS CLEAN, BUT MOST IS USEFUL"

Dean Abbott



Biological & technical artifact sources · Visual pattern recognition · Live inspection in MNE

What you should walk away with

Identify artifact sources

Recognize the major biological and technical sources of contamination in EEG recordings.

Distinguish by eye

Visually separate artifact types from genuine neural signal in raw traces.

Justify removal

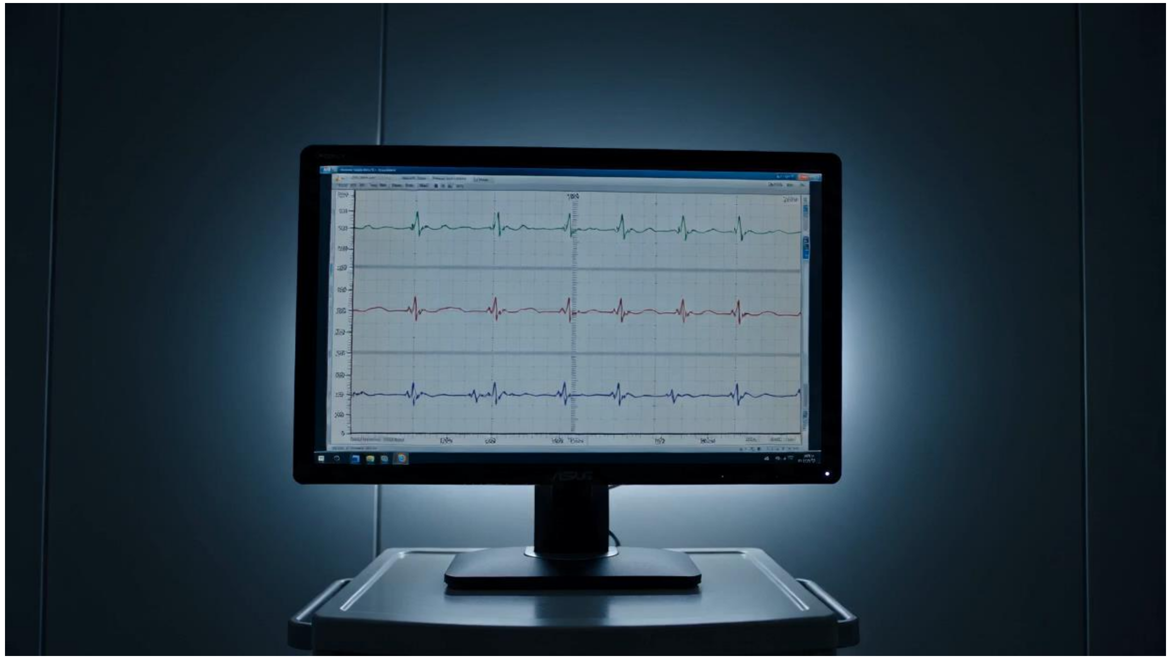
Explain why artifacts must be identified and handled before any downstream analysis.

What is Artifact

Artifacts are unwanted waves or recorded signals that are captured not due to brain activities

Artifacts may interfere with the amplitude, hide real neural activities, and lead to false conclusions.

It can be caused by physiologic, electric, or even environmental factors



Why ?

The possibility of having artifacts in an EEG is very high.

EEG electrodes measure all electrical activities not just from brain activities, but also from areas close to the scalp

Signal + Noise

=

EEG

Eye Movements

Muscle Activities

Brain Activities

Heartbeats

Electrode Problems

Three parts

Part A

Biological Artifacts

Ocular, muscle, cardiac, and respiration sources.

Part B

Technical / Environmental

Line noise, electrode pops, movement, amplifier noise.

Part C

Visual Inspection in MNE

Live demo, annotation, and a classification activity.

Biological Artifacts

Signal contamination generated by the participant’s own body.

OCULAR ARTIFACT · EOG

Eye Blinks & Eye Movements

Signature

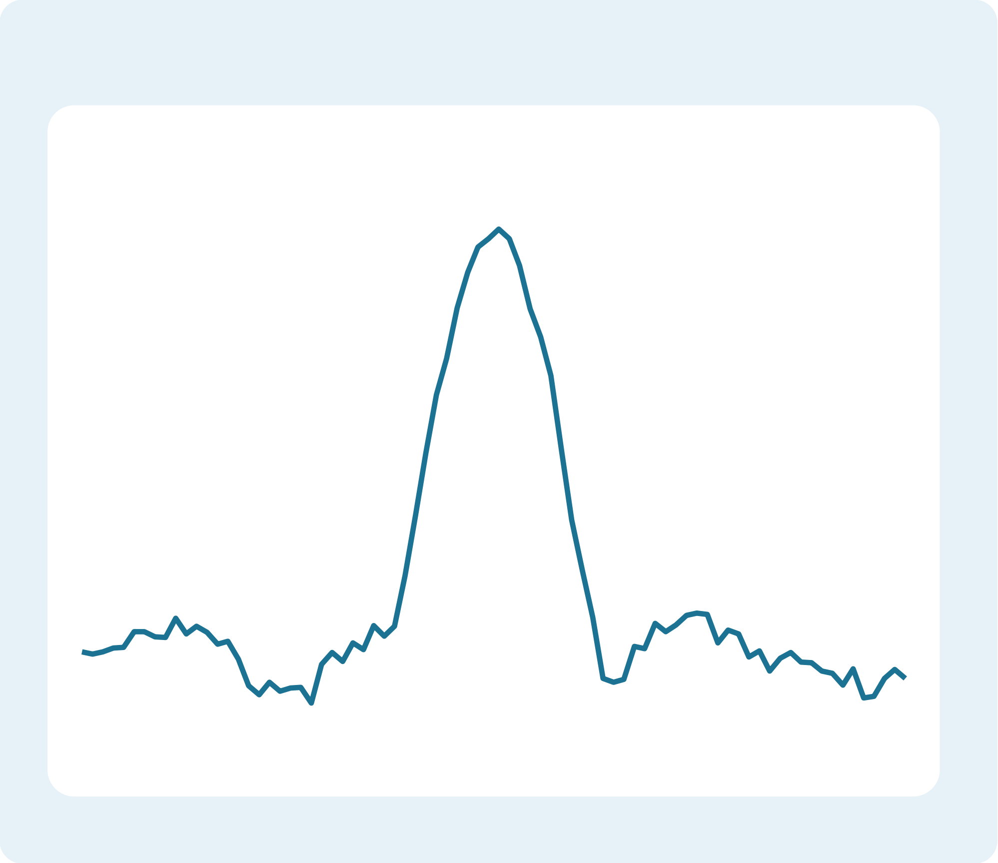
Blinks produce large, slow deflections up to about 200 µV. Eye movements instead create step-like shifts in the trace.

Where it shows up

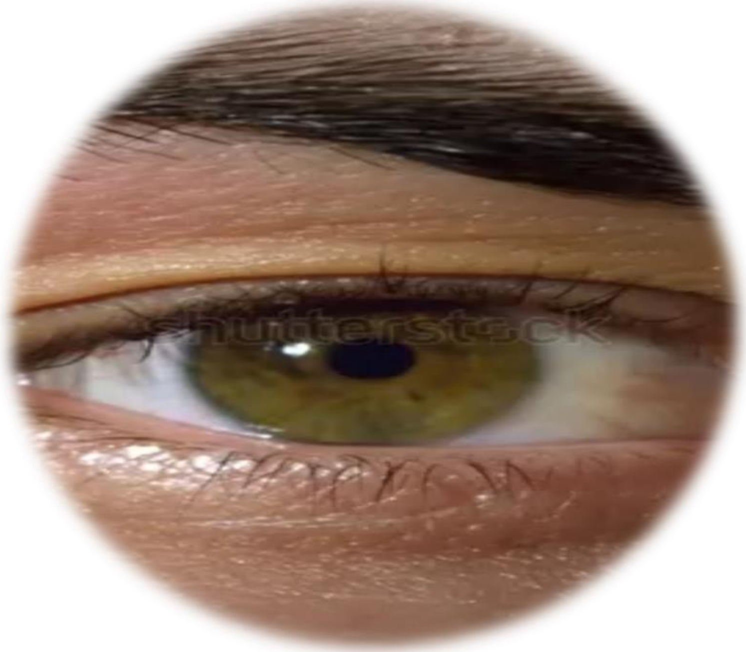
Most prominent at frontal electrodes, especially Fp1 and Fp2, closest to the eyes.

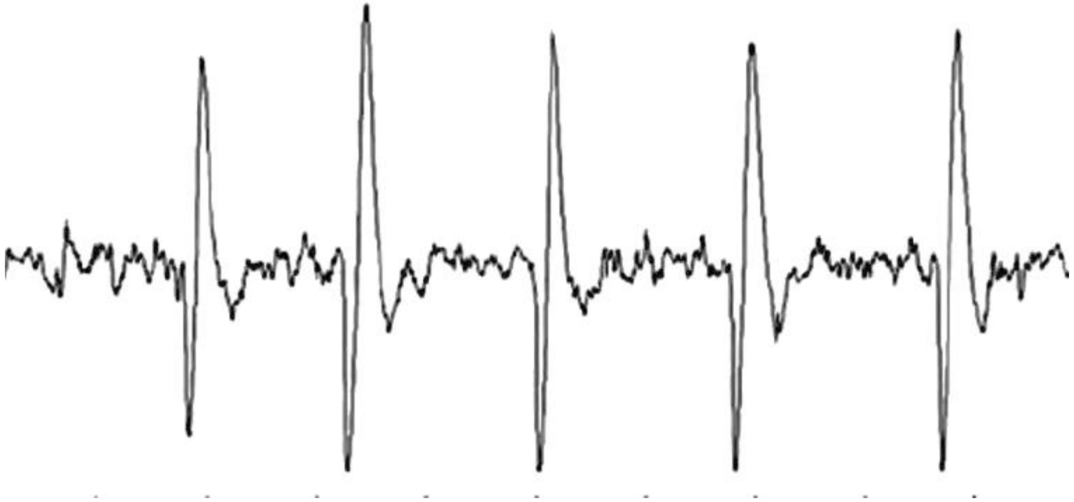
Typical cause

Movement of the corneo-retinal dipole during blinking or gaze shifts.



Eye Blinks & Eye Movements





Muscle Tension & Movement

Signature

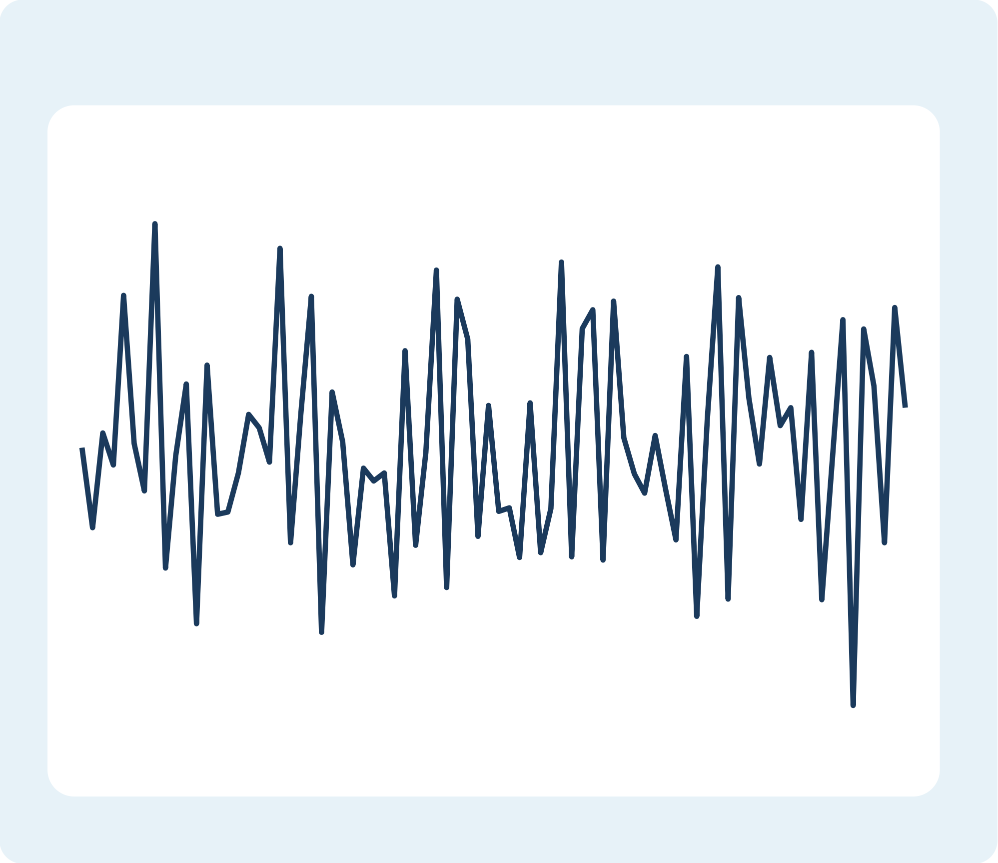
High-frequency, broadband noise above roughly 30 Hz that looks fuzzy and dense compared to neural rhythms.

Where it shows up

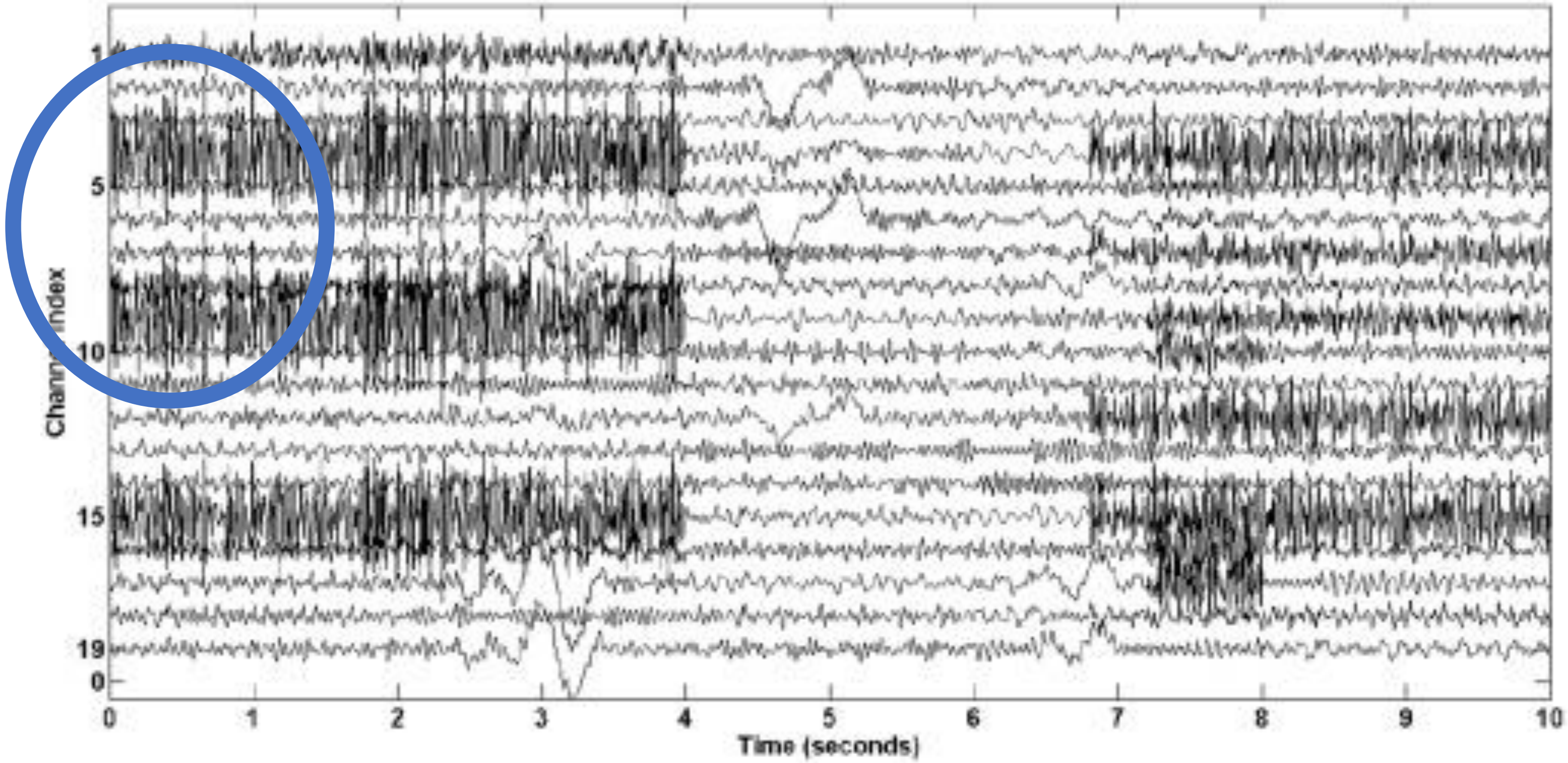
Most visible at temporal and frontal channels, nearest the jaw and forehead muscles.

Typical cause

Jaw clenching, facial tension, frowning, swallowing, or neck movement.



Muscle Tension & Movement



CARDIAC ARTIFACT · ECG / BCG

Heartbeat Contamination

Signature

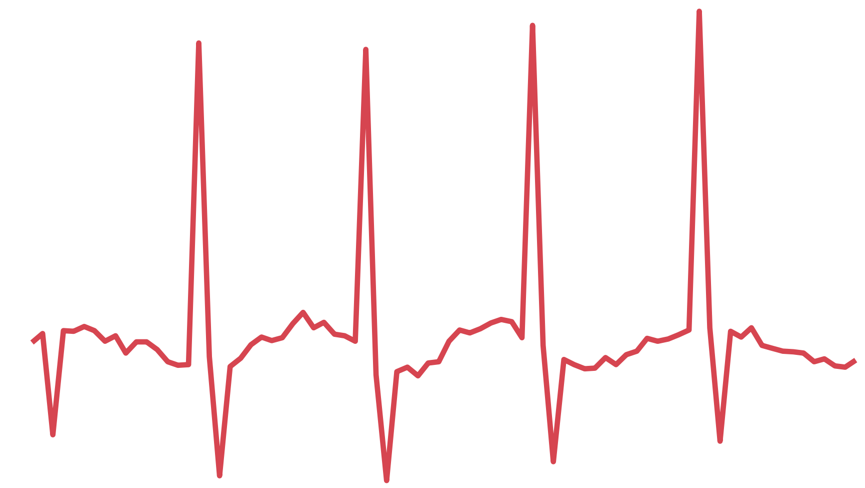
A regularly repeating QRS-like spike riding on top of the EEG, appearing at a steady heart-rate interval.

Where it shows up

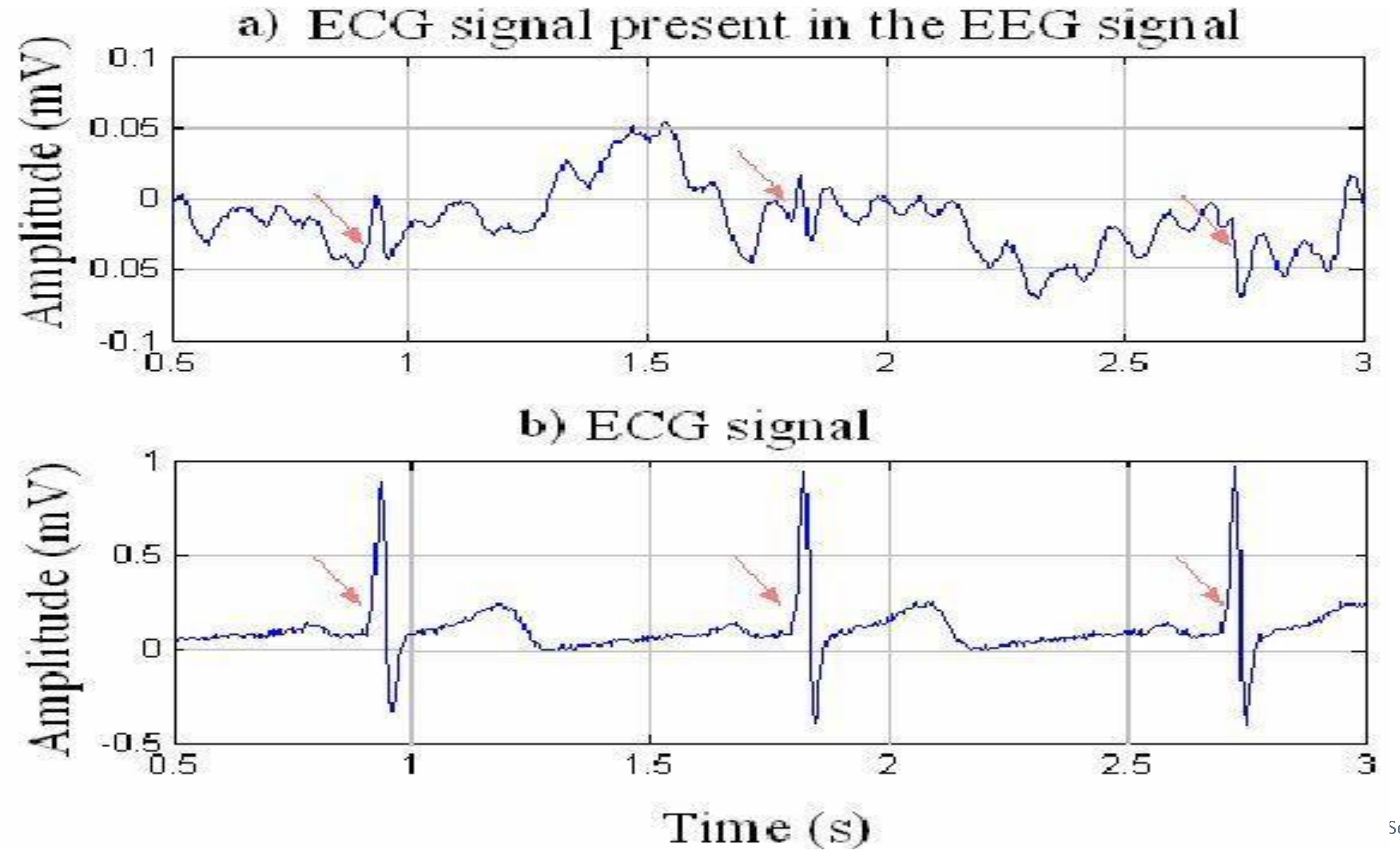
Can appear across many channels; strength depends on electrode placement and cable routing.

Typical cause

Electrical activity of the heart, more common in EEG-fMRI or supine (lying-down) recordings.



Heartbeat Contamination



RESPIRATION ARTIFACT

Breathing-Related Drift

Signature

A slow, smooth, rhythmic drift in the baseline that rises and falls with the breathing cycle.

Where it shows up

Seen broadly across channels, most obvious in long recordings.

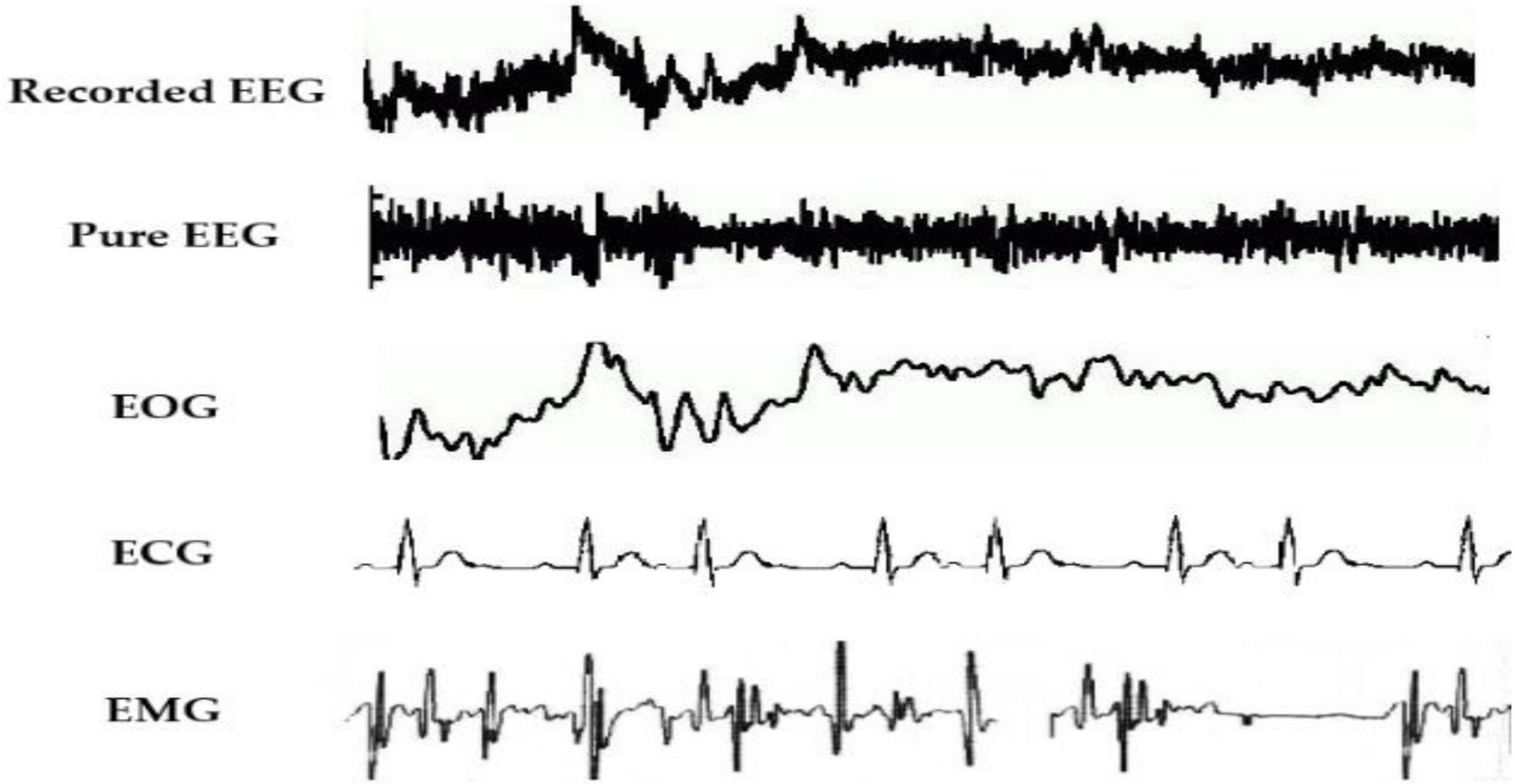
Typical cause

Chest and body movement associated with normal breathing.

PART A RECAP

Biological artifacts at a glance

Source	Visual signature	Where it's worst
Ocular (EOG)	Large slow deflections (blinks, up to ~200 µV) or step-like shifts (movements)	Frontal: Fp1, Fp2
Muscle (EMG)	High-frequency broadband noise, > 30 Hz	Temporal / frontal
Cardiac (ECG/BCG)	Regular QRS-like spikes at heart-rate interval	Widespread; posture-dependent
Respiration	Slow rhythmic baseline drift	Widespread, long recordings


Recorded EEG
Pure EEG
EOG
ECG
EMG

PART B

Technical & Environmental Artifacts

Contamination from equipment, wiring, and the recording environment.

LINE NOISE

Power-Line Interference

Signature

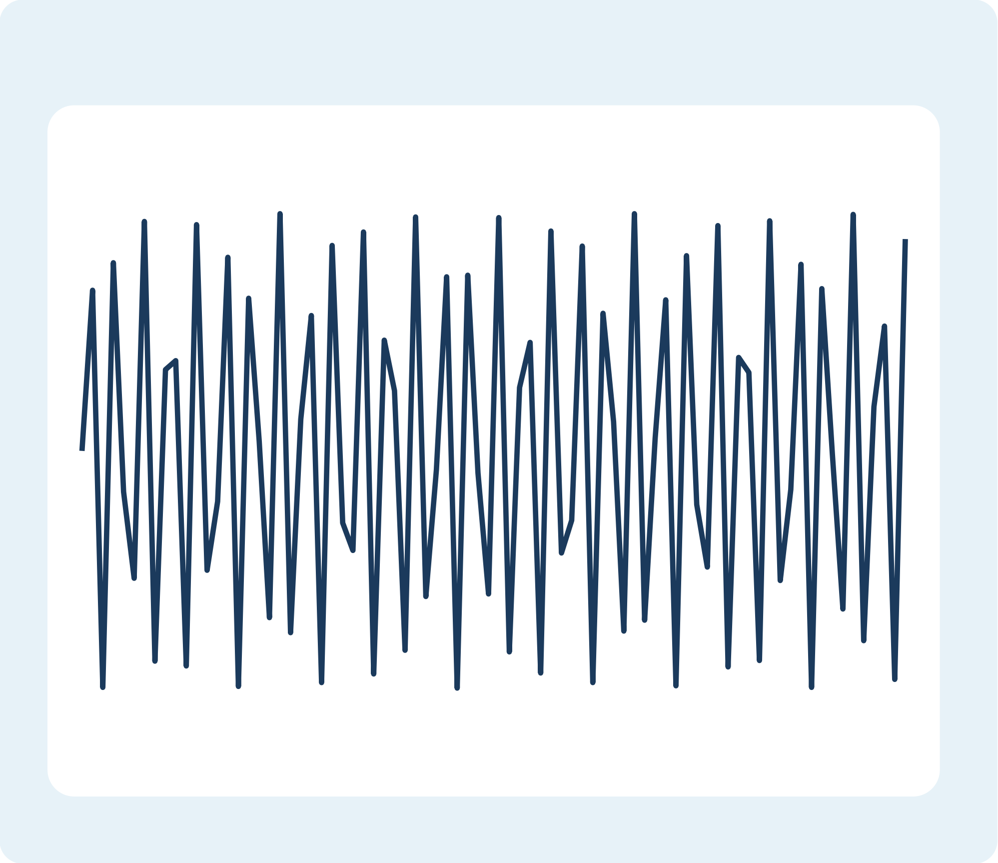
A very sharp, narrow peak in the power spectrum usually at 50 Hz.

Where it shows up

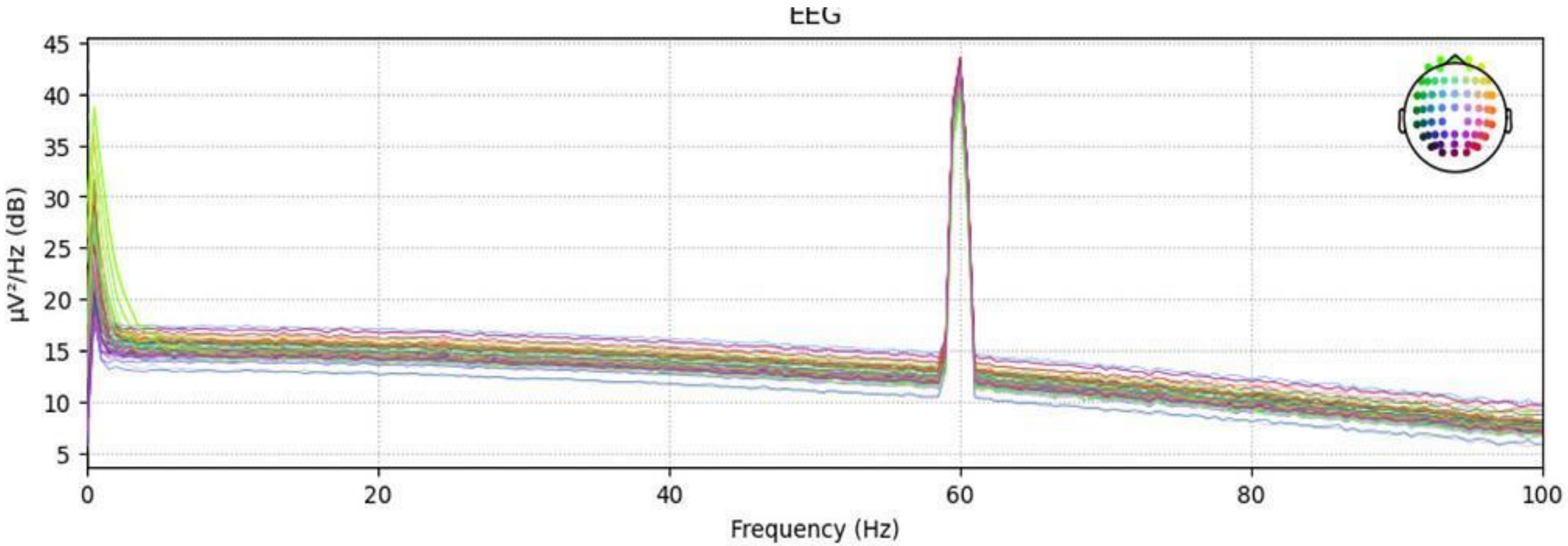
Affects all channels roughly equally since it is picked up from ambient electrical fields.

Typical cause

Nearby AC power lines, monitors, chargers, unshielded cables, or poor grounding.



Power-Line Interference



ELECTRODE POP

Sudden Single-Channel Spike

Signature

An abrupt, very large deflection confined to a single channel, unlike anything in the surrounding trace.

Where it shows up

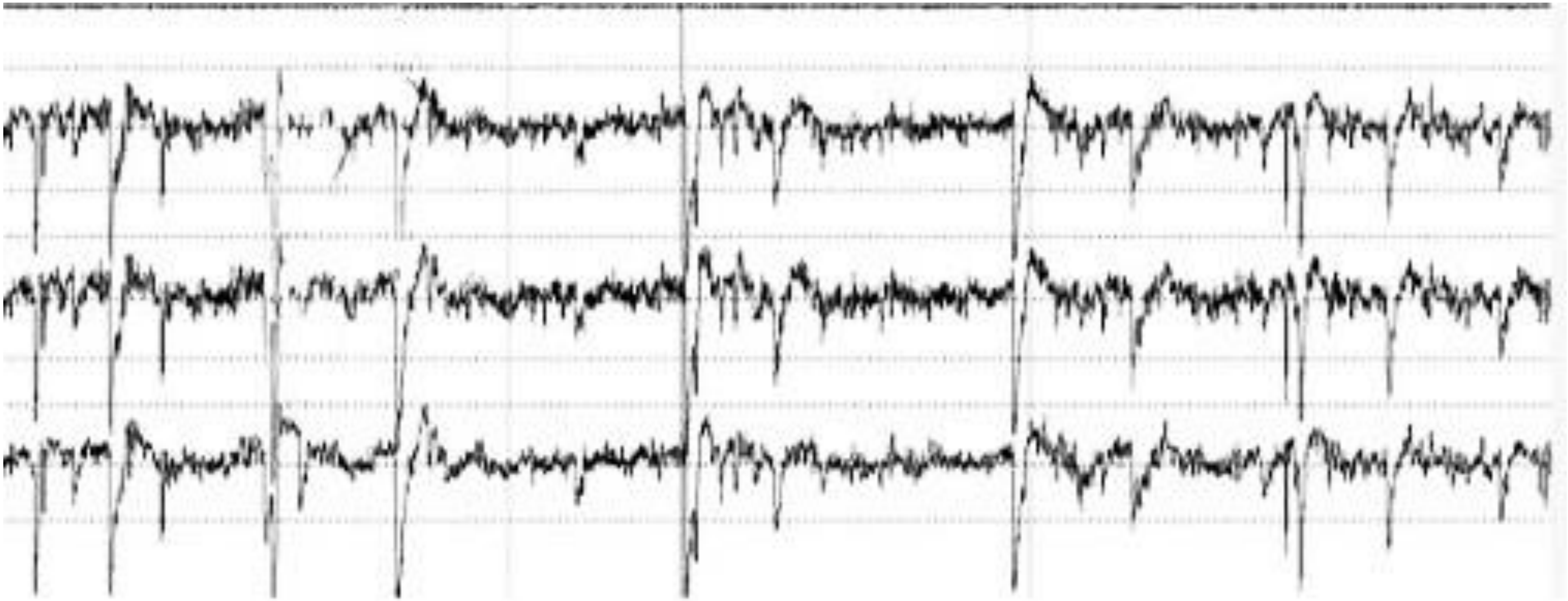
Localized to one electrode; neighboring channels stay clean.

Typical cause

Gel drying out, a loose connection, or a brief impedance change at that electrode.

ELECTRODE POP

Sudden Single-Channel Spike



MOVEMENT ARTIFACT

Large-Scale Body Movement

Signature

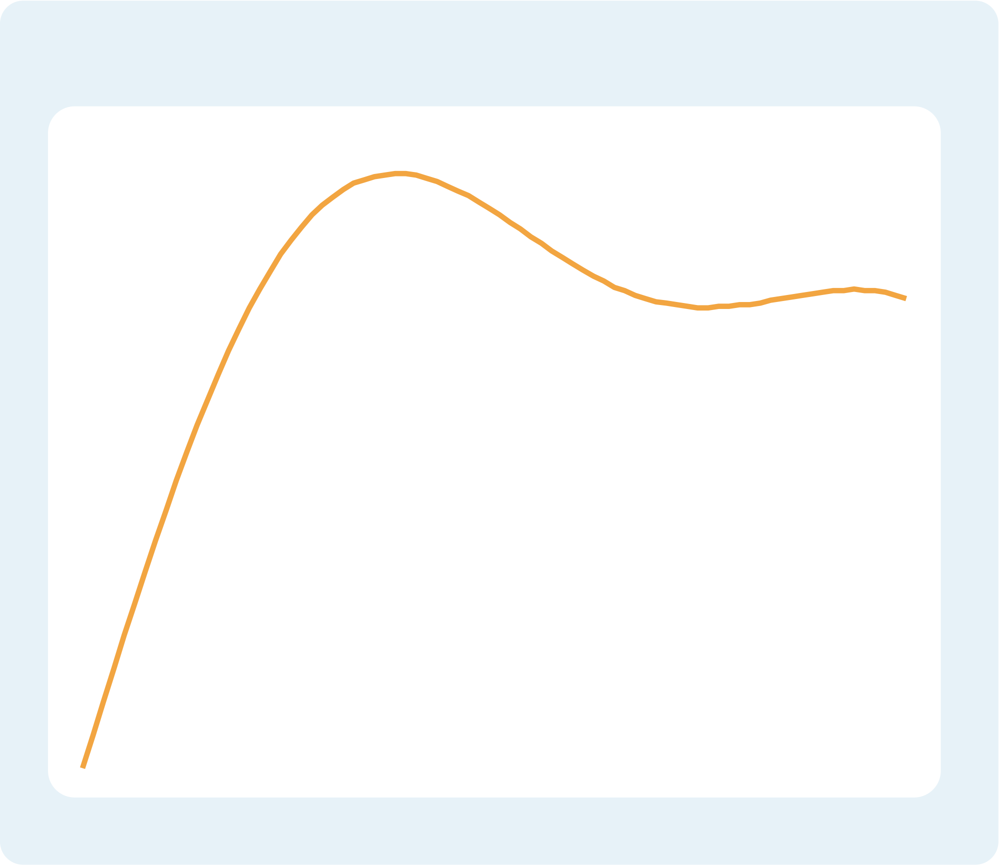
Slow, large-amplitude drifts that sweep across many channels at once, often irregular in shape.

Where it shows up

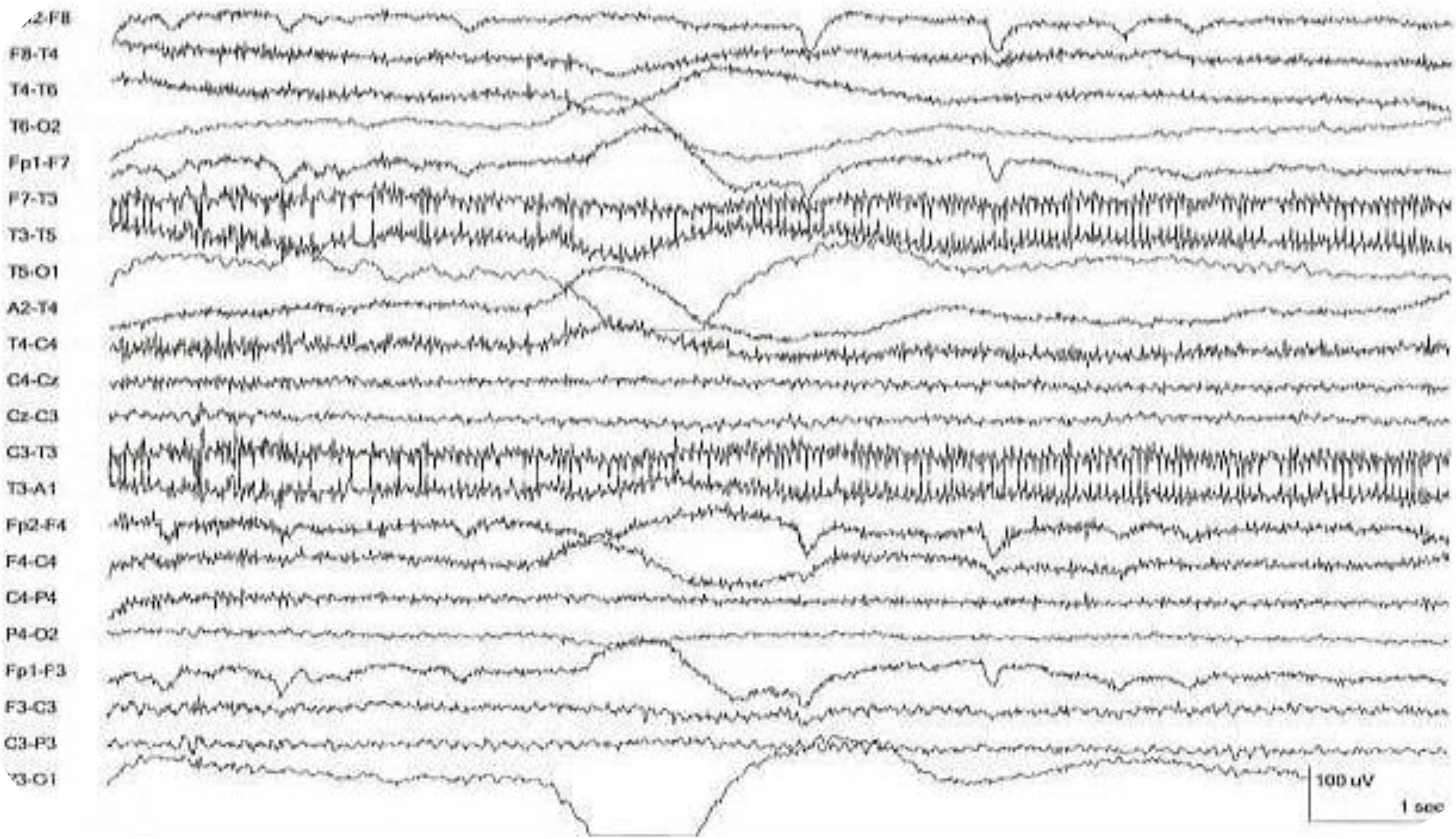
Broad, multi-channel involvement rather than one isolated electrode.

Typical cause

Head or body shifts, cable tugging, or postural adjustments during recording.



Large-Scale Body Movement



CABLE / AMPLIFIER NOISE

Broadband Noise Floor

Signature

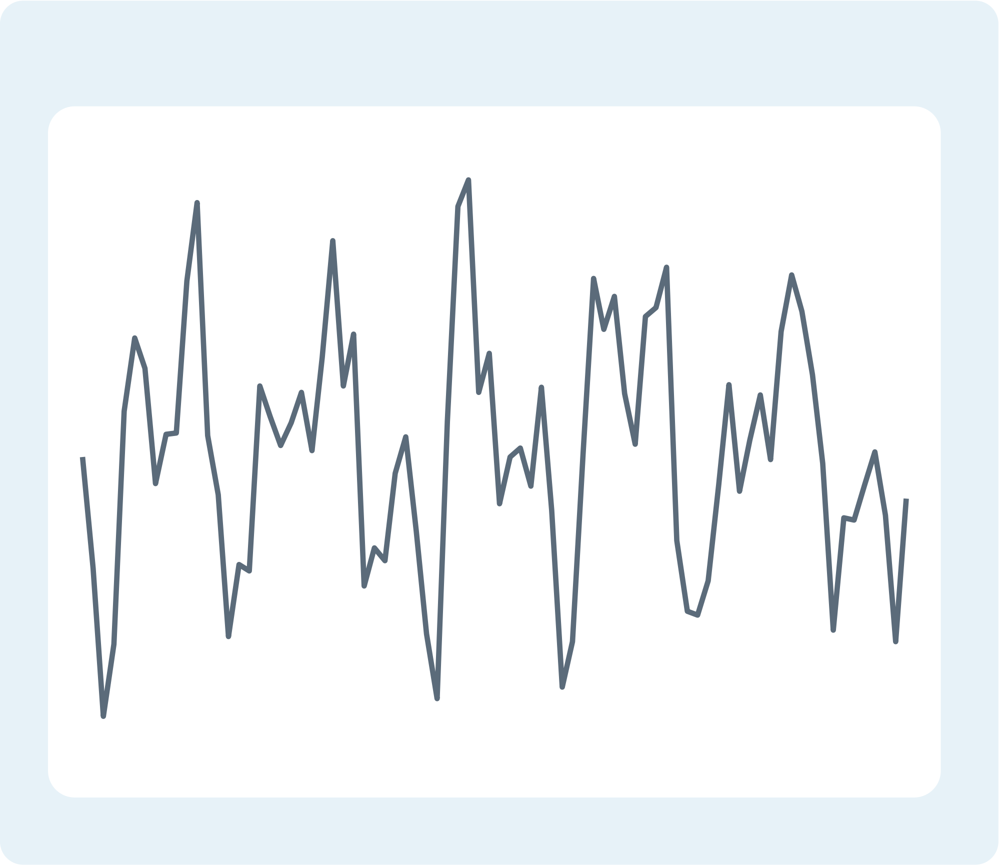
A persistent, low-level broadband noise floor with no clear rhythm or peak frequency.

Where it shows up

Can affect one channel (bad cable) or the whole montage (amplifier issue).

Typical cause

Damaged or poorly shielded cables, faulty connectors, or amplifier hardware issues.



PART B RECAP

Technical artifacts at a glance

Source	Visual signature	Scope
Line noise	Sharp 50/60 Hz peak in the power spectrum	All channels
Electrode pop	Sudden large single-channel spike	One channel
Movement	Slow, large-amplitude multi-channel drift	Many channels
Cable / amplifier noise	Persistent broadband noise floor	One channel or whole montage

PART C

Visual Inspection in MNE

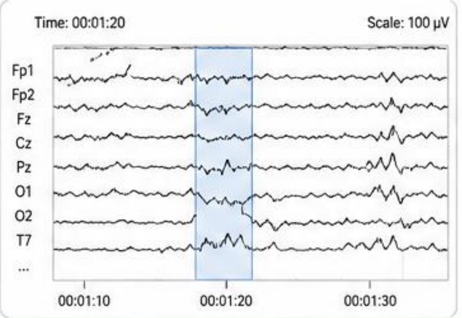
Automated methods are useful, BUT human inspection is very essential

Visual Inspection Workflow for EEG Data

1 Load EEG

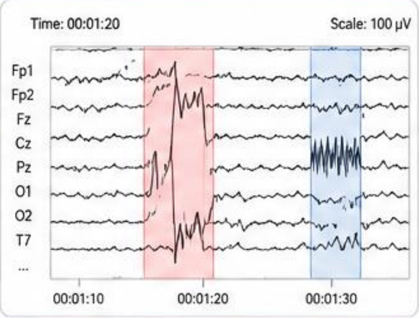
- Load the raw EEG file
- Check basic information (channels, duration, sampling rate)

2 Scroll through recording



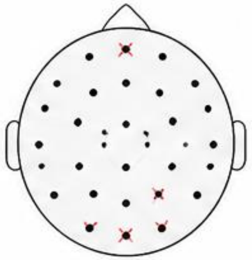
- Scroll through the entire recording
- Use zoom in/out to inspect the data closely

3 Identify artifacts



- Look for eye blinks, eye movements, muscle activity, line noise, electrode pops, drift, etc.
- Note time points and channels where artifacts occur

4 Mark bad channels

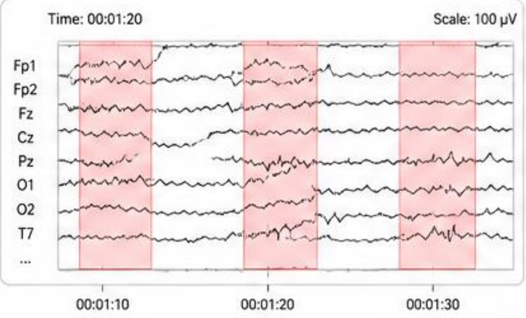


Bad channels

- Fp2
- T7
- O1

- Identify persistently noisy or flat channels
- Mark them as bad (exclude from analysis or interpolate)

5 Annotate contaminated segments



Annotations

- BAD_blink
- BAD_muscle
- BAD_noise
- BAD_channel

- Mark time intervals with artifacts using annotations
- Helps in later rejection or correction steps

6 Decide whether to reject or keep

Is the data suitable for analysis?

No → **Reject**
- Too much artifact
- Affects data quality
- Not recoverable

Yes → **Keep**
- Minimal artifact
- Recoverable
- Data still useful

↓ **Proceed to Preprocessing / Analysis**

- Reject segments or epochs if necessary
- Keep good data for further preprocessing and analysis

ACTIVITY

Classify the Waveform

Eye blink

Muscle

Cardiac

Line noise

Clean EEG

[Click here]

[Google Colab]

Why this matters

- Artifacts can dwarf the neural signal in amplitude — a single blink or muscle burst can swamp genuine EEG.
- Every artifact type has a recognizable signature: amplitude, frequency, shape, and spatial spread all give it away.
- Removing or correcting artifacts before analysis protects the validity of everything downstream — ERPs, spectral power, and connectivity measures alike.

Next session: automated artifact correction with ICA.

THANK YOU